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JEE MAIN · QUESTION PAPER

# JEE Main 2021 (Online) 17th March Evening Shift

JEE Main Question Paper with Answer Key

| 17 Mar 2021 | 90 | 20 | English |
|---|---|---|---|
| HELD ON | QUESTIONS | OUT OF SYLLABUS | LANGUAGE |

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## Chemistry 20 questions

**Q1.** MCQ (Single Correct Answer)

Open solution

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{Enzyme\ A} \underset{Glucose}{C_6H_{12}O_6} + \underset{Fructose}{C_6H_{12}O_6}$$

$$\underset{Glucose}{C_6H_{12}O_6} \xrightarrow{Enzyme\ B} 2C_2H_5OH + 2CO_2$$

In the above reactions, the enzyme A and enzyme B respectively are :

(A) Zymase and Invertase

(B) Invertase and Amylase

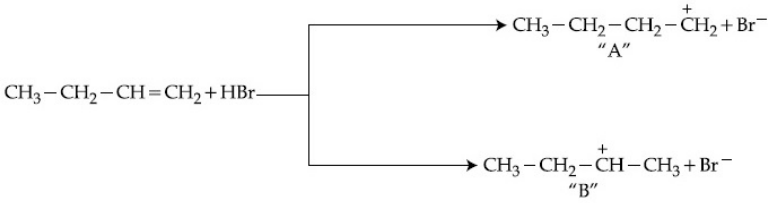
(C) Amylase and Invertase

(D) Invertase and Zymase

**Q2.** MCQ (Single Correct Answer)

Open solution

Choose the correct statement regarding the formation of carbocations A and B given.




(A) Carbocation A is more stable and formed relatively at faster rate

(B) Carbocation B is more stable and formed relatively at slow rate

(C) Carbocation B is more stable and formed relatively at faster rate

(D) Carbocation A is more stable and formed relatively at slow rate

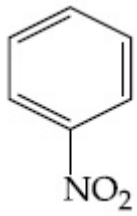
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**Q3.** MCQ (Single Correct Answer)

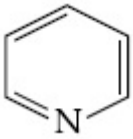
Open solution

Nitrogen can be estimated by Kjeldahl's method for which of the following compound?

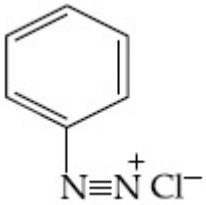
(A)



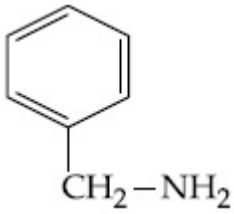
(B)



(C)



(D)



**Q4.** MCQ (Single Correct Answer)

Open solution

Amongst the following, the linear species is :

(A) $O_3$ (B) $Cl_2O$ (C) $N_3^-$ (D) $NO_2$

**Q5.** MCQ (Single Correct Answer)

Open solution

The correct pair(s) of the ambident nucleophiles is(are)

(A) AgCN/KCN

(B) RCOOAg/RCOOK

(C) $AgNO_2/KNO_2$

(D) AgI/KI

**(A)** (B) and (C) only **(B)** (B) only **(C)** (A) and (C) only **(D)** (A) only

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**Q6.** MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : 2-methylbutane on oxidation with $KMnO_4$ gives 2-methylbutan-2-ol.

Statement II : n-alkanes can be easily oxidised to corresponding alcohols with $KMnO_4$.

Choose the correct option :

**(A)** Both statement I and statement II are incorrect

**(B)** Both statement I and statement II are correct

**(C)** Statement I is correct but statement II is incorrect

**(D)** Statement I is incorrect but statement II is correct

**Q7.** MCQ (Single Correct Answer)

Open solution

During which of the following processes, does entropy decrease?

(A) Freezing of water to ice at $0^\circ C$

(B) Freezing of water to ice at $-10^\circ C$

(C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(D) Adsorption of CO(g) on lead surface.

(E) Dissolution of NaCl in water

Choose the correct answer from the options given below :

**(A)** (A), (C) and (E) only

**(B)** (B) and (C) only

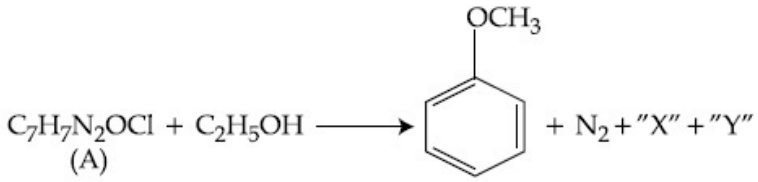
**(C)** (A), (B), (C) and (D) only

**(D)** (A) and (E) only

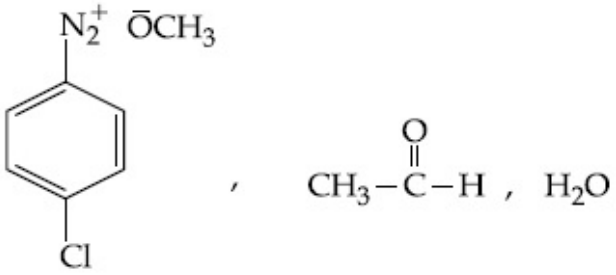
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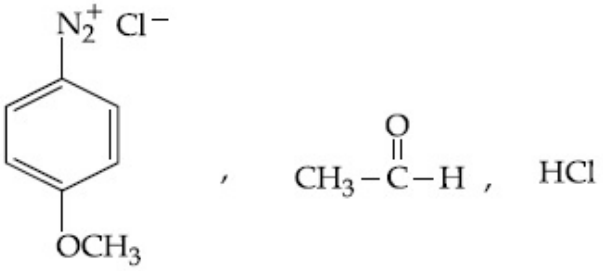
**Q8.** MCQ (Single Correct Answer)

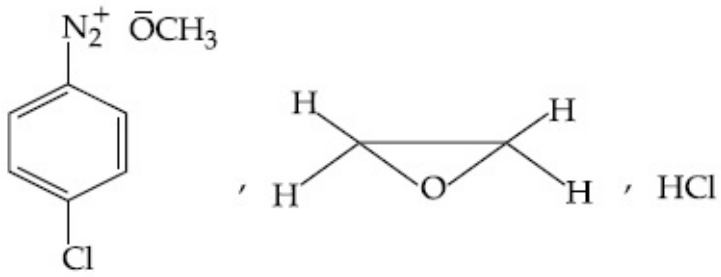
Open solution

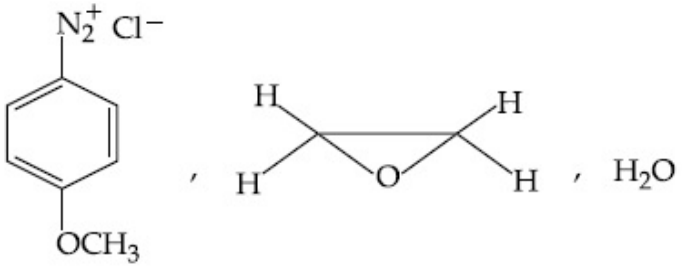



In the above reaction, the structural formula of (A), 'X' and 'Y' respectively are :

(A) 


(B) 


(C) 


(D) 


**Q9.** MCQ (Single Correct Answer)

Open solution

Primary, secondary and tertiary amines can be separated using:

(A) Chloroform and KOH

(B) Acetyl amide

(C) Benzene sulphonic acid

(D) para-Toluene sulphonyl chloride

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**Q10.** MCQ (Single Correct Answer)

Match List - I with List - II :

| | List - I | | List - II |
|---|---|---|---|
| (a) | $[Co(NH_3)_6][Cr(CN)_6]$ | (i) | Linkage isomerism |
| (b) | $[Co(NH_3)_3(NO_2)_3]$ | (ii) | Solvate isomerism |
| (c) | $[Cr(H_2O)_6]Cl_3$ | (iii) | Co-ordination isomerism |
| (d) | $cis-[CrCl_2(ox)_2]^{3-}$ | (iv) | Optical isomerism |

Choose the correct answer from the options given below :

**(A)** (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

**(B)** (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

**(C)** (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

**(D)** (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

Open solution

**Q11.** MCQ (Single Correct Answer)

Fructose is an example of :

**(A)** Heptose **(B)** Pyranose **(C)** Ketohexose **(D)** Aldohexose

Open solution

**Q12.** MCQ (Single Correct Answer)

The common positive oxidation states for an element with atomic number 24, are :

**(A)** +1 and +3 **(B)** +2 to +6 **(C)** +1 to +6 **(D)** +1 and +3 to +6

Open solution

**Q13.** Numerical

A KCl solution of conductivity 0.14 S $m^{-1}$ shows a resistance of 4.19$\Omega$ in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03$\Omega$. The conductivity of the HCl solution is _____ $\times 10^{-2}$ S $m^{-1}$. (Round off to the Nearest Integer).

Open solution

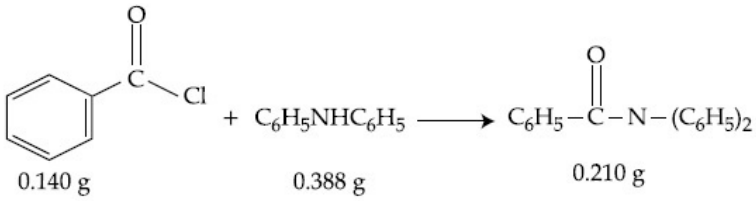
**Q14.** Numerical

On complete reaction of $FeCl_3$ with oxalic acid in aqueous solution containing KOH, resulted in the formation of product A. The secondary valency of Fe in the product A is _____. (Round off to the Nearest Integer).

Open solution

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**Q15.** Numerical




Open solution

Consider the above reaction. The percentage yield of amide product is _____. (Round off to the Nearest Integer).

(Given : Atomic mass : C : 12.0 u, H : 1.0 u, N : 14.0 u, O : 16.0 u, Cl : 35.5 u)

**Q16.** Numerical

The reaction 2A + $B_2 \rightarrow$ 2AB is an elementary reaction.

For a certain quantity of reactants, if the volume of the reaction vessel is reduced by a factor of 3, the rate of the reaction increases by a factor of _____. (Round off to the Nearest Integer).

Open solution

**Q17.** Numerical

Consider the reaction

$$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$$

The temperature at which $K_C$ = 20.4 and $K_P$ = 600.1, is _____ K. (Round off to the Nearest Integer). [Assume all gases are ideal and R = 0.0831 L bar $K^{-1}$ $mol^{-1}$]

Open solution

**Q18.** Numerical

In the ground state of atomic Fe(Z = 26), the spin-only magnetic moment is _____ $\times 10^{-1}$ BM. (Round off to the Nearest Integer). [Given : $\sqrt{3}$ = 1.73, $\sqrt{2}$ = 1.41 ]

Open solution

**Q19.** Numerical

A 1 molal $K_4Fe(CN)_6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another solution which contains 18.1 weight percent of a non electrolytic solute A. The molar mass of A is _____ u. (Round off to the Nearest Integer). [Density of water = 1.0 g $cm^{-3}$ ]

Open solution

**Q20.** Numerical

The total number of C-C sigma bond/s in mesityl oxide ($C_6H_{10}O$) is _____. (Round off to the Nearest Integer).

Open solution

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## Mathematics 22 questions

**Q21.** MCQ (Single Correct Answer)

If the sides AB, BC and CA of a triangle ABC have 3, 5 and 6 interior points respectively, then the total number of triangles that can be constructed using these points as vertices, is equal to :

(A) 240 (B) 360 (C) 333 (D) 364

Open solution

**Q22.** MCQ (Single Correct Answer)

Let a computer program generate only the digits 0 and 1 to form a string of binary numbers with probability of occurrence of 0 at even places be $\frac{1}{2}$ and probability of occurrence of 0 at the odd place be $\frac{1}{3}$. Then the probability that '10' is followed by '01' is equal to :

(A) $\frac{1}{18}$ (B) $\frac{1}{3}$ (C) $\frac{1}{9}$ (D) $\frac{1}{6}$

Open solution

**Q23.** MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as f(x) = $e^{-x}$sinx. If F : [0, 1] $\rightarrow$ R is a differentiable function with that F(x) = $\int_0^x f(t)dt$, then the value of $\int_0^1 (F'(x) + f(x))e^x dx$ lies in the interval

(A) $\left[\frac{331}{360}, \frac{334}{360}\right]$ (B) $\left[\frac{330}{360}, \frac{331}{360}\right]$

(C) $\left[\frac{335}{360}, \frac{336}{360}\right]$ (D) $\left[\frac{327}{360}, \frac{329}{360}\right]$

Open solution

**Q24.** MCQ (Single Correct Answer)

Let $S_1$, $S_2$ and $S_3$ be three sets defined as

$S_1 = \{z \in C : |z - 1| \leq \sqrt{2}\}$

$S_2 = \{z \in C : Re((1 - i)z) \geq 1\}$

$S_3 = \{z \in C : Im(z) \leq 1\}$

Then the set $S_1 \cap S_2 \cap S_3$ :

(A) has exactly three elements (B) is a singleton

(C) has infinitely many elements (D) has exactly two elements

Open solution

**Q25.** MCQ (Single Correct Answer)

The number of solutions of the equation

$\sin^{-1}\left[x^2 + \frac{1}{3}\right] + \cos^{-1}\left[x^2 - \frac{2}{3}\right] = x^2$, for $x \in [-1, 1]$, and [x] denotes the greatest integer less than or equal to x, is :

(A) 0 (B) Infinite (C) 2 (D) 4

Open solution

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**Q26.** MCQ (Single Correct Answer)

Open solution

If the curve y = y(x) is the solution of the differential equation

$2(x^2+x^{5/4})dy-y(x+x^{1/4})dx=2x^{9/4}dx$, x > 0 which

passes through the point $\left(1,1-\frac{4}{3}\log_e 2\right)$, then the value of y(16) is equal to :

(A) $4\left(\frac{31}{3}-\frac{8}{3}\log_e 3\right)$ (B) $\left(\frac{31}{3}-\frac{8}{3}\log_e 3\right)$

(C) $\left(\frac{31}{3}+\frac{8}{3}\log_e 3\right)$ (D) $4\left(\frac{31}{3}+\frac{8}{3}\log_e 3\right)$

**Q27.** MCQ (Single Correct Answer)

Open solution

Let O be the origin. Let $\overrightarrow{OP}=x\hat{i}+y\hat{j}-\hat{k}$ and $\overrightarrow{OQ}=-\hat{i}+2\hat{j}+3x\hat{k}$, x, y∈R, x > 0, be such that $\left|\overrightarrow{PQ}\right|=\sqrt{20}$ and the vector $\overrightarrow{OP}$ is perpendicular $\overrightarrow{OQ}$. If $\overrightarrow{OR}=3\hat{i}+z\hat{j}-7\hat{k}$, z∈R, is coplanar with $\overrightarrow{OP}$ and $\overrightarrow{OQ}$, then the value of $x^2+y^2+z^2$ is equal to :

(A) 2 (B) 9 (C) 7 (D) 1

**Q28.** MCQ (Single Correct Answer)

Open solution

If the integral

$\int_0^{10}\frac{[\sin 2\pi x]}{e^{x-[x]}}dx=\alpha e^{-1}+\beta e^{-\frac{1}{2}}+\gamma$, where $\alpha,\beta,\gamma$ are integers and [x] denotes the greatest integer less than or equal to x, then the value of $\alpha+\beta+\gamma$ is equal to :

(A) 0 (B) 10 (C) 20 (D) 25

**Q29.** MCQ (Single Correct Answer)

Open solution

The value of the limit

$\lim\limits_{\theta\to 0}\frac{\tan(\pi\cos^2\theta)}{\sin(2\pi\sin^2\theta)}$ is equal to :

(A) 0 (B) $-\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

**Q30.** MCQ (Single Correct Answer)

Open solution

Consider the function f : R → R defined by

$f(x)=\begin{cases}\left(2-\sin\left(\frac{1}{x}\right)\right)|x|, & x\neq 0\\ 0, & x=0\end{cases}$. Then f is :

(A) not monotonic on $(-\infty, 0)$ and $(0, \infty)$

(B) monotonic on $(0, \infty)$ only

(C) monotonic on $(-\infty, 0)$ only

(D) monotonic on $(-\infty, 0)\cup(0, \infty)$

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**Q31.** MCQ (Single Correct Answer)

The value of $\lim\limits_{n\to\infty}\frac{[r]+[2r]+...+[nr]}{n^2}$, where r is a non-zero real number and [r] denotes the greatest integer less than or equal to r, is equal to :

(A) r (B) $\frac{r}{2}$ (C) 0 (D) 2r

Open solution

**Q32.** MCQ (Single Correct Answer)

If x, y, z are in arithmetic progression with common difference d, $x \neq 3d$, and the determinant of the matrix $\begin{bmatrix}3 & 4\sqrt{2} & x\\ 4 & 5\sqrt{2} & y\\ 5 & k & z\end{bmatrix}$ is zero, then the value of $k^2$ is :

(A) 72 (B) 12 (C) 36 (D) 6

Open solution

**Q33.** MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential equation

$\cos x(3\sin x+\cos x+3)dy=(1+y\sin x(3\sin x+\cos x+3))dx, 0\le x\le\frac{\pi}{2}, y(0)=0$. Then, $y\left(\frac{\pi}{3}\right)$ is equal to :

(A) $2\log_e\left(\frac{\sqrt{3}+7}{2}\right)$ (B) $2\log_e\left(\frac{3\sqrt{3}-8}{4}\right)$

(C) $2\log_e\left(\frac{2\sqrt{3}+10}{11}\right)$ (D) $2\log_e\left(\frac{2\sqrt{3}+9}{6}\right)$

Open solution

**Q34.** Numerical

Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ and $B=\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\neq\begin{bmatrix}0\\ 0\end{bmatrix}$ such that AB = B and a + d = 2021, then the value of $ad-bc$ is equal to _____.

Open solution

**Q35.** Numerical

Let the coefficients of third, fourth and fifth terms in the expansion of $\left(x+\frac{a}{x^2}\right)^n, x\neq 0$, be in the ratio 12 : 8 : 3. Then the term independent of x in the expansion, is equal to _____.

Open solution

**Q36.** Numerical

Let $\tan\alpha$, $\tan\beta$ and $\tan\gamma$; $\alpha,\beta,\gamma\neq\frac{(2n-1)\pi}{2}$, $n\in N$ be the slopes of three line segments OA, OB and OC, respectively, where O is origin. If circumcentre of $\Delta ABC$ coincides with origin and its orthocentre lies on y-axis, then the value of $\left(\frac{\cos 3\alpha+\cos 3\beta+\cos 3\gamma}{\cos\alpha\cos\beta\cos\gamma}\right)^2$ is equal to _____.

Open solution

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**Q37.** Numerical

Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $\left(3\hat{i} + 2\hat{j} - \hat{k}\right)$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to _____.

Open solution

**Q38.** Numerical

Let $I_n = \int_1^e x^{19}(\log|x|)^n dx$, where n∈N. If (20)$I_{10}$ = $\alpha I_9$ + $\beta I_8$, for natural numbers $\alpha$ and $\beta$, then $\alpha - \beta$ equals to _____.

Open solution

**Q39.** Numerical

If 1, $\log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x, then the value of the determinant $\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x - 1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ is equal to :

Open solution

**Q40.** Numerical

Consider a set of 3n numbers having variance 4. In this set, the mean of first 2n numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first 2n numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then 9k is equal to _____.

Open solution

**Q41.** Numerical

Let f : $[-1, 1] \to$ R be defined as f(x) = $ax^2$ + bx + c for all x∈$[-1, 1]$, where a, b, c∈R such that f$(-1)$ = 2, f'$(-1)$ = 1 for x∈$(-1, 1)$ the maximum value of f "(x) is $\frac{1}{2}$. If f(x) $\leq \alpha$, x∈$[-1, 1]$, then the least value of $\alpha$ is equal to _____.

Open solution

**Q42.** Numerical

Let f : $[-3, 1] \to$ R be given as

$$f(x) = \begin{cases} \min\{(x+6), x^2\}, -3 \leq x \leq 0 \\ \max\{\sqrt{x}, x^2\}, \ 0 \leq x \leq 1. \end{cases}$$

If the area bounded by y = f(x) and x-axis is A, then the value of 6A is equal to _____.

Open solution

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## Physics 28 questions

**Q43.** MCQ (Single Correct Answer)

A sound wave of frequency 245 Hz travels with the speed of 300 $ms^{-1}$ along the positive x-axis. Each point of the wave moves to and fro through a total distance of 6 cm. What will be the mathematical expression of this travelling wave?

Open solution

(A) $Y(x, t) = 0.03 [\sin 5.1x - (0.2 \times 10^3)t]$

(B) $Y(x, t) = 0.03 [\sin 5.1x - (1.5 \times 10^3)t]$

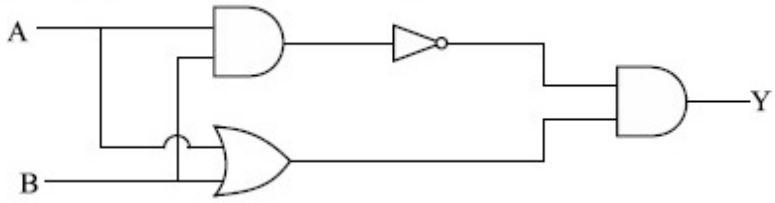
(C) $Y(x, t) = 0.06 [\sin 5.1x - (1.5 \times 10^3)t]$

(D) $Y(x, t) = 0.06 [\sin 0.8x - (0.5 \times 10^3)t]$

**Q44.** MCQ (Single Correct Answer)

Which one of the following will be the output of the given circuit?

Open solution




(A) XOR Gate (B) NOR Gate (C) NAND Gate (D) AND Gate

**Q45.** MCQ (Single Correct Answer)

Match List - I with List - II

Open solution

| | List - I | | List - II |
|---|---|---|---|
| (a) | Phase difference between current and voltage in a purely resistive AC circuit | (i) | $\frac{\pi}{2}$; current leads voltage |
| (b) | Phase difference between current and voltage in a pure inductive AC circuit | (ii) | zero |
| (c) | Phase difference between current and voltage in a pure capacitive AC circuit | (iii) | $\frac{\pi}{2}$; current lags voltage |
| (d) | Phase difference between current and voltage in an LCR series circuit | (iv) | $\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$ |

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(D) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

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**Q46.** MCQ (Single Correct Answer)

The atomic hydrogen emits a line spectrum consisting of various series. Which series of hydrogen atomic spectra is lying in the visible region?

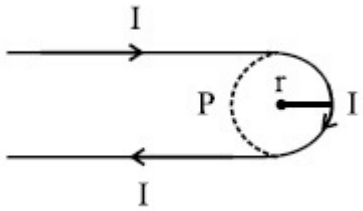
(A) Brackett series (B) Balmer series (C) Paschen series (D) Lyman series

Open solution

**Q47.** MCQ (Single Correct Answer)

A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?

Open solution




(A) $\frac{\mu_0 I}{4\pi r}(2-\pi)$ (B) $\frac{\mu_0 I}{2\pi r}(2-\pi)$

(C) $\frac{\mu_0 I}{4\pi r}(2+\pi)$ (D) $\frac{\mu_0 I}{2\pi r}(2+\pi)$

**Q48.** MCQ (Single Correct Answer)

Two identical photocathodes receive the light of frequencies $f_1$ and $f_2$ respectively. If the velocities of the photo-electrons coming out are $v_1$ and $v_2$ respectively, then

Open solution

(A) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{\frac{1}{2}}$ (B) $v_1^2 + v_2^2 = \frac{2h}{m}[f_1 + f_2]$

(C) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{\frac{1}{2}}$ (D) $v_1^2 - v_2^2 = \frac{2h}{m}[f_1 - f_2]$

**Q49.** MCQ (Single Correct Answer)

If one mole of the polyatomic gas is having two vibrational modes and $\beta$ is the ratio of molar specific heats for polyatomic gas $\left(\beta = \frac{C_P}{C_V}\right)$ then the value of $\beta$ is :

Open solution

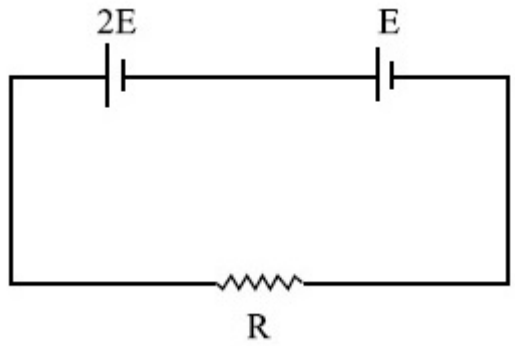
(A) 1.02 (B) 1.35 (C) 1.2 (D) 1.25

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**Q50.** MCQ (Single Correct Answer)

Two cells of emf 2E and E with internal resistance $r_1$ and $r_2$ respectively are connected in series to an external resistor R (see figure). The value of R, at which the potential difference across the terminals of the first cell becomes zero is

Open solution




(A) $r_1 - r_2$ (B) $\frac{r_1}{2} - r_2$ (C) $\frac{r_1}{2} + r_2$ (D) $r_1 + r_2$

**Q51.** MCQ (Single Correct Answer)

What happens to the inductive reactance and the current in a purely inductive circuit if the frequency is halved?

Open solution

(A) Both, inducting reactance and current will be doubled.

(B) Inductive reactance will be doubled and current will be halved.

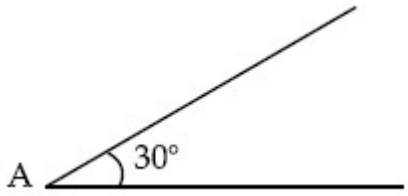
(C) Both, inductive reactance and current will be halved.

(D) Inductive reactance will be halved and current will be doubled.

**Q52.** MCQ (Single Correct Answer)

A sphere of mass 2 kg and radius 0.5 m is rolling with an initial speed of 1 $ms^{-1}$ goes up an inclined plane which makes an angle of 30° with the horizontal plane, without slipping. How long will the sphere take to return to the starting point A?

Open solution




(A) 0.60 s (B) 0.52 s (C) 0.80 s (D) 0.57 s

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**Q53.** MCQ (Single Correct Answer)

Two particles A and B of equal masses are suspended from two massless springs of spring constants $K_1$ and $K_2$ respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A and B is

Open solution

(A) $\frac{K_1}{K_2}$ (B) $\sqrt{\frac{K_1}{K_2}}$ (C) $\frac{K_2}{K_1}$ (D) $\sqrt{\frac{K_2}{K_1}}$

**Q54.** MCQ (Single Correct Answer)

The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is x = 0 at t = 0; then its displacement after time (t = 1) is :

Open solution

(A) $v_0 + g + f$ (B) $v_0 + \frac{g}{2} + \frac{F}{3}$

(C) $v_0 + 2g + 3F$ (D) $v_0 + \frac{g}{2} + F$

**Q55.** MCQ (Single Correct Answer)

A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball. (Take g = 10 ms$^{-2}$)

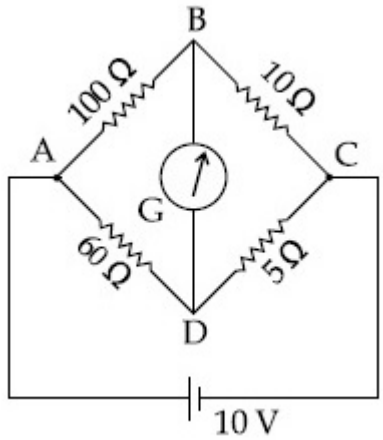
Open solution

(A) 2.50 ms$^{-1}$ (B) 3.0 ms$^{-1}$ (C) 2.0 ms$^{-1}$ (D) 3.50 ms$^{-1}$

**Q56.** MCQ (Single Correct Answer)

The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10V is maintained across AC.

Open solution




(A) 4.87 $\mu$A (B) 4.87 mA (C) 2.44 mA (D) 2.44 $\mu$A

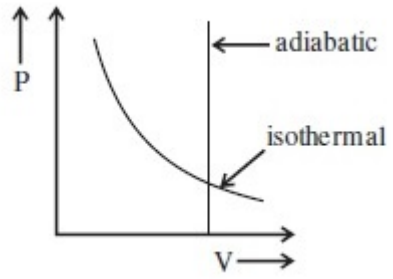
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**Q57.** MCQ (Single Correct Answer)

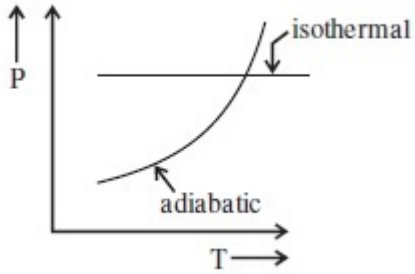
Open solution

Which one is the correct option for the two different thermodynamic processes?

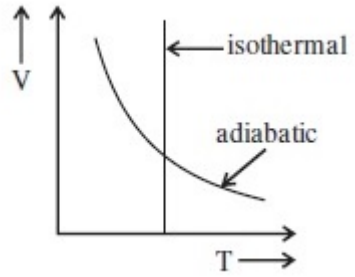
(a)



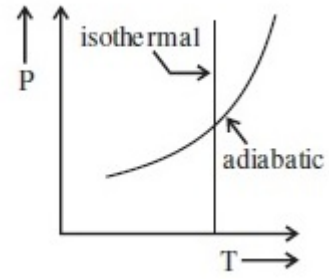

(b)




(c)




(d)



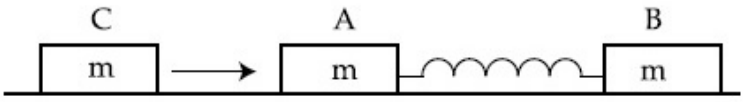

**(A)** (a) only **(B)** (c) and (d) **(C)** (b) and (c) **(D)** (c) and (a)

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**Q58.** MCQ (Single Correct Answer)

Two identical blocks A and B each of mass m resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant K. A third block C of mass m moving with a speed v along the line joining A and B collides with A. The maximum compression in the spring is

Open solution




(A) $\sqrt{\frac{mv}{K}}$ (B) $\sqrt{\frac{m}{2K}}$ (C) $\sqrt{\frac{mv}{2K}}$ (D) $v\sqrt{\frac{m}{2K}}$

**Q59.** MCQ (Single Correct Answer)

A geostationary satellite is orbiting around an arbitrary planet 'P' at a height of 11R above the surface of 'P', R being the radius of 'P'. The time period of another satellite in hours at a height of 2R from the surface of 'P' is _____. 'P' has the time period of 24 hours.

Open solution

(A) 3 (B) 5 (C) $6\sqrt{2}$ (D) $\frac{6}{\sqrt{2}}$

**Q60.** MCQ (Single Correct Answer)

An object is located at 2 km beneath the surface of the water. If the fractional compression $\frac{\Delta V}{V}$ is 1.36%, the ratio of hydraulic stress to the corresponding hydraulic strain will be _____. [Given : density of water is 1000 kgm$^{-3}$ and g = 9.8 ms$^{-2}$]

Open solution

(A) $1.44 \times 10^7$ Nm$^{-2}$ (B) $1.44 \times 10^9$ Nm$^{-2}$

(C) $1.96 \times 10^7$ Nm$^{-2}$ (D) $2.26 \times 10^9$ Nm$^{-2}$

**Q61.** Numerical

Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm$^3$ of oleic acid per cm$^3$ of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm$^2$ by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3}$ cm. Then the thickness of oleic acid layer will be x $\times 10^{-14}$ m. Where x is _____.

Open solution

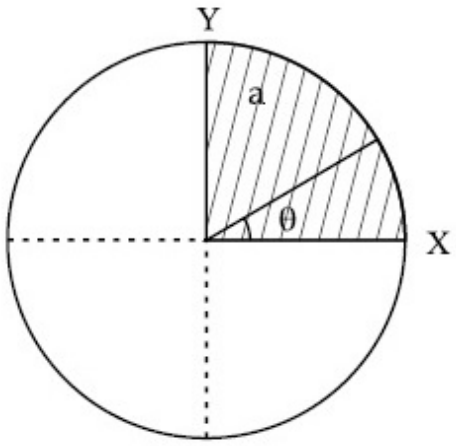
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**Q62.** Numerical

The disc of mass M with uniform surface mass density $\sigma$ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is _____. (Round off to the Nearest Integer).

Open solution

[a is an area as shown in the figure]




**Q63.** Numerical

The image of an object placed in air formed by a convex refracting surface is at a distance of 10 m behind the surface. The image is real and is at $\frac{2}{3}^{rd}$ of the distance of the object from the surface. The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$ m. The value of 'x' is _____.

Open solution

**Q64.** Numerical

The electric field intensity produced by the radiation coming from a 100 W bulb at a distance of 3 m is E. The electric field intensity produced by the radiation coming from 60W at the same distance is $\sqrt{\frac{x}{5}}$E. Where the value of x = _____.

Open solution

**Q65.** Numerical

The electric field in a region is given by $\overrightarrow{E} = \frac{2}{5}E_0\hat{i} + \frac{3}{5}E_0\hat{j}$ with $E_0 = 4.0 \times 10^3 \frac{N}{C}$. The flux of this field through a rectangular surface area 0.4 $m^2$ parallel to the Y-Z plane is _____ $Nm^2C^{-1}$.

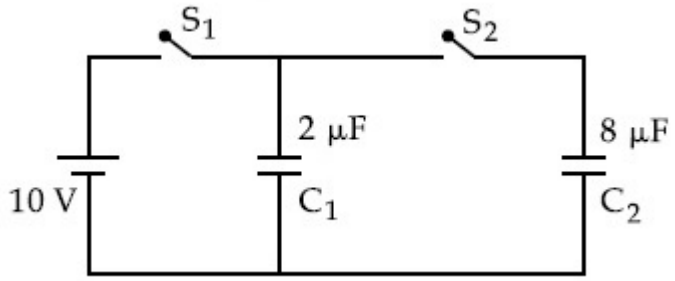
Open solution

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**Q66.** Numerical

A $2\mu F$ capacitor $C_1$ is first charged to a potential difference of 10V using a battery. Then the battery is removed and the capacitor is connected to an uncharged capacitor $C_2$ of 8 $\mu F$. The charge in $C_2$ on equilibrium condition is _____ $\mu C$. (Round off to the Nearest Integer)

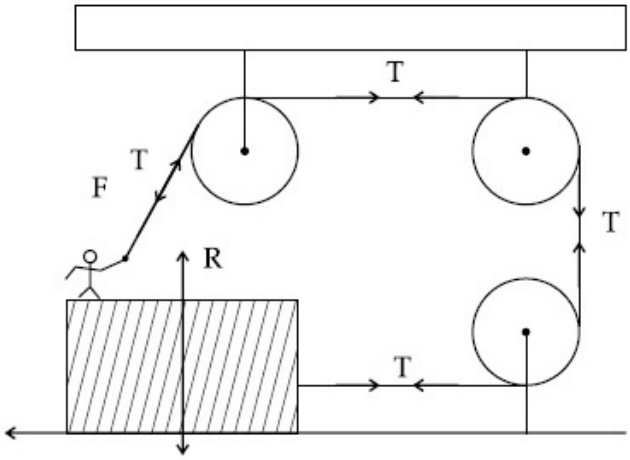
Open solution




**Q67.** Numerical

A boy of mass 4 kg is standing on a piece of wood having mass 5 kg. If the coefficient of friction between the wood and the floor is 0.5, the maximum force that the boy can exert on the rope so that the piece of wood does not move from its place is _____ N. (Round off to the Nearest Integer) [Take g = 10 $ms^{-2}$ ]

Open solution




**Q68.** Numerical

A body of mass 1 kg rests on a horizontal floor with which it has a coefficient of static friction $\frac{1}{\sqrt{3}}$. It is desired to make the body move by applying the minimum possible force F N. The value of F will be _____. (Round off to the Nearest Integer) [Take g = 10 $ms^{-2}$ ]

Open solution

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**Q69.** Numerical

A particle of mass m moves in a circular orbit in a central potential field U(r) = $U_0r^4$. If Bohr's quantization conditions are applied, radii of possible orbitals $r_n$ vary with $n^{\frac{1}{\alpha}}$, where $\alpha$ is _____.

Open solution

**Q70.** Numerical

Seawater at a frequency f = $9 \times 10^2$ Hz, has permittivity $\varepsilon = 80\varepsilon_0$ and resistivity $\rho = 0.25\ \Omega$m. Imagine a parallel plate capacitor is immersed in seawater and is driven by an alternating voltage source V(t) = $V_0$ sin(2πft). Then the conduction current density becomes $10^x$ times the displacement current density after time t = $\frac{1}{800}$s. The value of x is _____. (Given : $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ $Nm^2C^{-2}$)

Open solution

## Out of syllabus 20 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

**Q71.** MCQ (Single Correct Answer) · **Chemistry**

The set of elements that differ in mutual relationship from those of the other sets is :

(A) Li - Na (B) Be - Al (C) B - Si (D) Li - Mg

Open solution

**Q72.** MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

| | List - I | | List - II |
|---|---|---|---|
| (a) | Haematite | (i) | $Al_2O_3 \,.\, xH_2O$ |
| (b) | Bauxite | (ii) | $Fe_2O_3$ |
| (c) | Magnetite | (iii) | $CuCO_3 \,.\, Cu(OH)_2$ |
| (d) | Malachite | (iv) | $Fe_3O_4$ |

Choose the correct answer from the options given below :

(A) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(C) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(D) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

Open solution

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**Q73.** MCQ (Single Correct Answer) · **Chemistry**

The set that represents the pair of neutral oxides of nitrogen is :

(A) NO and $N_2O$ (B) $N_2O$ and $NO_2$ (C) NO and $NO_2$ (D) $N_2O$ and $N_2O_3$

Open solution

**Q74.** MCQ (Single Correct Answer) · **Chemistry**

For the coagulation of a negative sol, the species below, that has the highest flocculating power is :

(A) $Na^+$ (B) $Ba^{2+}$ (C) $PO_4^{3-}$ (D) $SO_4^{2-}$

Open solution

**Q75.** MCQ (Single Correct Answer) · **Chemistry**

Which of the following statement(s) is (are) incorrect reason for eutrophication?

(A) excess usage of fertilisers

(B) excess usage of detergents

(C) dense plant population in water bodies

(D) lack of nutrients in water bodies that prevent plant growth

Choose the most appropriate answer from the options given below :

(A) (A) Only (B) (B) and (D) only (C) (C) only (D) (D) only

Open solution

**Q76.** MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II.

| | **List - I Chemical Compound** | | **List - II Used as** |
|---|---|---|---|
| (a) | Sucralose | (i) | Synthetic detergent |
| (b) | Glyceryl ester of stearic acid | (ii) | Artificial sweetener |
| (c) | Sodium benzoate | (iii) | Antiseptic |
| (d) | Bithionol | (iv) | Food preservative |

Choose the correct match :

(A) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii) (B) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) (D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Open solution

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**Q77.** MCQ (Single Correct Answer) · **Chemistry**

The functional groups that are responsible for the ion-exchange property of cation and anion exchange resins, respectively, are :

(A) $-SO_3H$ and $-NH_2$

(B) $-NH_2$ and COOH

(C) $-NH_2$ and $-SO_3H$

(D) $-SO_3H$ and $-COOH$

Open solution

**Q78.** MCQ (Single Correct Answer) · **Chemistry**

One of the by-products formed during the recovery of $NH_3$ from Solvay process is :

(A) $Ca(OH)_2$

(B) $NH_4Cl$

(C) $NaHCO_3$

(D) $CaCl_2$

Open solution

**Q79.** Numerical · **Chemistry**

The number of chlorine atoms in 20 mL of chlorine gas at STP is _____ $10^{21}$. (Round off to the Nearest Integer).

[Assume chlorine is an ideal gas at STP

R = 0.083 L bar $mol^{-1}$ $K^{-1}$, $N_A = 6.023 \times 10^{23}$]

Open solution

**Q80.** Numerical · **Chemistry**

KBr is doped with $10^{-5}$ mole percent of $SrBr_2$. The number of cationic vacancies in 1g of KBr crystal is _____ $10^{14}$. (Round off to the Nearest Integer).

[Atomic Mass : K : 39.1 u, Br : 79.9 u $N_A = 6.023 \times 10^{23}$]

Open solution

**Q81.** MCQ (Single Correct Answer) · **Mathematics**

The number of solutions of the equation $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is :

(A) 4

(B) 3

(C) 2

(D) 5

Open solution

**Q82.** MCQ (Single Correct Answer) · **Mathematics**

Let the tangent to the circle $x^2 + y^2 = 25$ at the point R(3, 4) meet x-axis and y-axis at points P and Q, respectively. If r is the radius of the circle passing through the origin O and having centre at the incentre of the triangle OPQ, then $r^2$ is equal to :

(A) $\frac{585}{66}$

(B) $\frac{625}{72}$

(C) $\frac{529}{64}$

(D) $\frac{125}{72}$

Open solution

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**Q83.** MCQ (Single Correct Answer) · **Mathematics**

If the equation of plane passing through the mirror image of a point (2, 3, 1) with respect to line $\frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$ and containing the line $\frac{x-2}{3} = \frac{1-y}{2} = \frac{z+1}{1}$ is $\alpha x + \beta y + \gamma z = 24$, then $\alpha + \beta + \gamma$ is equal to :

(A) 21 (B) 19 (C) 18 (D) 20

Open solution

**Q84.** MCQ (Single Correct Answer) · **Mathematics**

Two tangents are drawn from a point P to the circle $x^2 + y^2 - 2x - 4y + 4 = 0$, such that the angle between these tangents is $\tan^{-1}\left(\frac{12}{5}\right)$, where $\tan^{-1}\left(\frac{12}{5}\right) \in (0, \pi)$. If the centre of the circle is denoted by C and these tangents touch the circle at points A and B, then the ratio of the areas of $\Delta PAB$ and $\Delta CAB$ is :

(A) 3 : 1 (B) 9 : 4 (C) 2 : 1 (D) 11 : 4

Open solution

**Q85.** MCQ (Single Correct Answer) · **Mathematics**

If the Boolean expression $(p \wedge q) \odot (p \otimes q)$ is a tautology, then $\odot$ and $\otimes$ are respectively given by :

(A) $\vee, \rightarrow$

(B) $\rightarrow, \rightarrow$

(C) $\wedge, \vee$

(D) $\wedge, \rightarrow$

Open solution

**Q86.** MCQ (Single Correct Answer) · **Mathematics**

Let L be a tangent line to the parabola $y^2 = 4x - 20$ at (6, 2). If L is also a tangent to the ellipse $\frac{x^2}{2} + \frac{y^2}{b} = 1$, then the value of b is equal to :

(A) 20 (B) 14 (C) 16 (D) 11

Open solution

**Q87.** MCQ (Single Correct Answer) · **Mathematics**

The value of $\sum_{r=0}^{6} \left({}^6C_r \cdot {}^6C_{6-r}\right)$ is equal to :

(A) 924 (B) 1024 (C) 1124 (D) 1324

Open solution

**Q88.** Numerical · **Mathematics**

Let P be an arbitrary point having sum of the squares of the distances from the planes x + y + z = 0, $lx - nz = 0$ and $x - 2y + z = 0$, equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of $l - n$ is equal to _____.

Open solution

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**Q89.** MCQ (Single Correct Answer) · **Physics**

A carrier signal C(t) = 25 sin(2.512 $\times$ $10^{10}$t) is amplitude modulated by a message signal m(t) = 5 sin(1.57 $\times$ $10^{8}$t) and transmitted through an antenna. What will be the bandwidth of the modulated signal?

Open solution

(A) 50 MHz (B) 8 GHz (C) 1987.5 MHz (D) 2.01 GHz

**Q90.** MCQ (Single Correct Answer) · **Physics**

A block of mass 1 kg attached to a spring is made to oscillate with an initial amplitude of 12 cm. After 2 minutes the amplitude decreases to 6 cm. Determine the value of the damping constant for this motion . (take ln 2 = 0.693)

Open solution

(A) $0.69 \times 10^{2}$ kg $s^{-1}$ (B) $3.3 \times 10^{2}$ kg $s^{-1}$

(C) $1.16 \times 10^{-2}$ kg $s^{-1}$ (D) $5.7 \times 10^{-3}$ kg $s^{-1}$

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# Answer Key

## Chemistry

| | | | | |
|---|---|---|---|---|
| 1. **D** | 2. **C** | 3. **D** | 4. **C** | 5. **C** |
| 6. **C** | 7. **C** | 8. **B** | 9. **D** | 10. **B** |
| 11. **C** | 12. **B** | 13. **57** | 14. **6** | 15. **77** |
| 16. **27** | 17. **354** | 18. **49** | 19. **85** | 20. **5** |

## Mathematics

| | | | | |
|---|---|---|---|---|
| 21. **C** | 22. **C** | 23. **B** | 24. **C** | 25. **A** |
| 26. **A** | 27. **B** | 28. **A** | 29. **B** | 30. **A** |
| 31. **B** | 32. **A** | 33. **C** | 34. **2020** | 35. **4** |
| 36. **144** | 37. **486** | 38. **1** | 39. **2** | 40. **68** |
| 41. **5** | 42. **41** | | | |

## Physics

| | | | | |
|---|---|---|---|---|
| 43. **B** | 44. **A** | 45. **A** | 46. **B** | 47. **C** |
| 48. **D** | 49. **C** | 50. **B** | 51. **D** | 52. **D** |
| 53. **D** | 54. **B** | 55. **A** | 56. **B** | 57. **B** |
| 58. **D** | 59. **A** | 60. **B** | 61. **25** | 62. **4** |
| 63. **30** | 64. **3** | 65. **640** | 66. **16** | 67. **30** |
| 68. **5** | 69. **3** | 70. **6** | | |

## Out of syllabus

| | | | | |
|---|---|---|---|---|
| 71. **A** | 72. **B** | 73. **A** | 74. **B** | 75. **D** |
| 76. **D** | 77. **A** | 78. **D** | 79. **1** | 80. **5** |
| 81. **B** | 82. **B** | 83. **B** | 84. **B** | 85. **B** |
| 86. **B** | 87. **A** | 88. **0** | 89. **A** | 90. **C** |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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